Brandon Imstepf

Ph.D. Candidate
Applied Mathematics Program
University of California, Merced

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Research Interests

Mathematical Biology, Mathematical and Computational Neuroscience, Bioinformatics, Machine Learning, Data Science

Teaching Interests

Probability & Statistics, Calculus, Mathematical Modeling, Introductory Programming for Scientists (Python, R)

Education

Anticipated Summer 2027 — **University of California, Merced**, *Doctor of Philosophy*, Applied Mathematics
Doctoral Advisor: Dr. Suzanne Sindi

Spring 2020 — **University of California, Santa Cruz**, *Bachelor of Arts*, Mathematics
Concentration: Computational

Teaching Experience

Guest Lecturer, University of California, Merced

Fall 2026 — **Probability Theory & Statistics (MATH 032)**
Delivered three lectures to ∼200 students in place of the instructor of record, covering random variables, discrete distributions, and midterm review.

Teaching Assistant, University of California, Merced

Summer 2026, Fall 2026 — **Probability Theory & Statistics (MATH 032)**
Led 3 discussion sections (∼100 students); lectured on material, wrote and administer quizzes.

Summer 2024 — **Probability Theory & Statistics (MATH 032)**
Led 2 discussion sections (∼60 students), lectured on material, wrote and administered quizzes.

Fall 2024 — **Calculus I (MATH 021)**
Led 2 discussion sections (∼60 students), lectured on material, wrote and administered quizzes.

Tutoring

2018 — **Mathematics Tutor**, *Santa Monica College*
Tutored students in courses ranging from pre-algebra through calculus.

Research Experience

Summer 2022-Present **Graduate Student Researcher**, *University of California, Merced*
Research on mathematical models of neurodegenerative disease in the Sindi Lab.

Summer 2024-Present **Sindi-Raj Lab Collaboration**, *University of California, Merced - University of California, San Francisco*
Collaboration between the Sindi Lab and Ashish Raj's group (Department of Radiology, UCSF) modeling the transport of pathological tau protein in Alzheimer's Disease.

May 2025-August 2026 **Research Affiliate, Joint Genome Institute**, *Lawrence Berkeley National Lab*
Summer internship extended through the New Energy Science Workforce (RENEW) Initiative. Designed and implemented neural networks in Java for gene calling in BBTools (mentor: Brian Bushnell), outperforming deterministic approaches by at least 10%.

Presentations

October 2026 **Accelerating the Network Transport Model of Tau Protein Spread with Machine Learning**
Contributed Talk, Soceity of Industrial and Applied Mathematics, Northern and Central California Section 2026 (SIAM-NCC 2026), Davis, California

July 2026 **Machine Learning Accelerated Modeling of Tau Protein Spread with the Network Transport Model**
Contributed Talk, European Conference on Mathematical and Theoretical Biology (ECMTB'26), Graz, Austria

July 2026 **Machine Learning for Gene Calling: Neural Network Integration for BBTools' CallGenes**
Invited Presentation, European Conference on Mathematical and Theoretical Biology (ECMTB'26), Graz, Austria

July 2025 **Accelerating Solutions of Nonlinear PDEs Using Machine Learning: A Case Study with the Network Transport Model**
Contributed Talk, Society for Mathematical Biology 2025, Edmonton, Canada

July 2025 **Machine Learning Applications in Gene Annotation**
Research Presentation, Lawrence Berkeley National Lab

October 2024 **Bridging the Scales: From Protein to Brain in Alzheimer's Disease**
Invited Presentation, Society for Industrial and Applied Mathematics, Northern and Central California Section 2024

July 2024 **Bridging the Scales: From Protein to Brain in Alzheimer's Disease**
Invited Presentation, Korean Society for Mathematical Biology – Society for Mathematical Biology 2024, Konkuk University, Seoul

Sep 2023 **Predicting Atrophy in Alzheimer's Via Network Laplacian**
Invited Presentation, Mathematical Biology Seminar, University of California, Merced

Feb 2023 **A First Year's Perspective on Mathematical Modelling**
Invited Presentation, University of California, Merced

Honors & Awards

Summer 2026 **Landahl Travel Award**, *Society for Mathematical Biology*

Fall 2024, Fall 2025 **Research and Training Grant Recipient**, *UC Merced Graduate Division*, University of California, Merced

Fall 2023,2025, Spring 2024,2026, Summer 2024 — **Distinguished Service in Support of Applied Mathematics**, *UC Merced Graduate Division*, University of California, Merced

Summer 2022-Summer 2024 — **Graduate Research Fellowship**, *National Institutes of Health G-RISE Interdisciplinary Biomedical Science and Technology Program*, University of California, Merced

Summer 2022 — **Competitive Edge Summer Bridge Program Participant**, *UC Merced Graduate Division*, University of California, Merced

Service & Leadership

Fall 2023-Present — **Applied Math Graduate Student Leadership Committee**, *UC Merced Graduate Division*, University of California, Merced

Meet monthly with fellow graduate students and the graduate program chair and vice chair to address student concerns, departmental activities, and program direction.

Fall 2023-Fall 2024 — **School of Natural Sciences Graduate Student Council**, *UC Merced Graduate Division*, University of California, Merced

Met each semester with the Dean of Natural Sciences and representatives from other programs to advise on campus-wide activities and the future direction of the School.

Skills

Programming	Python, MATLAB, R, Java, C++, SQL
Scientific Computing	NumPy, SciPy, PyTorch
Tools	LaTeX, Git, Jupyter, Quarto/RMarkdown, HPC/Slurm
Teaching	CatCourses (Canvas), Gradescope

Memberships

Society for Mathematical Biology

Society for Industrial and Applied Mathematics

American Mathematical Society

References

Suzanne Sindi, PhD, *University of California, Merced*, Professor, Applied Math, Doctoral Advisor
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Ashish Raj, PhD, *University of California, San Francisco*, Professor, Primary Collaborator
Ashish.Raj@ucsf.edu

Keith Thompson, *University of California, Merced*, Continuing Lecturer, Teaching Reference
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